Krisna Joko Purjianto

Software Engineer

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Surakarta, Central Java, Indonesia (Open to relocate)

SUMMARY

Software Engineer focused on backend development using PHP/Laravel and Golang. Experienced in API development, database management (MySQL & PostgreSQL), payment gateway integration, and server setup & deployment. Capable of working across the stack with modern frontend frameworks (Vue.js, React.js, Next.js). My ability to unravel complex problems, simplify the findings, and draw clear conclusions has become my advantage in technical problem solving and improving system efficiency. I take full ownership in my role, enjoy collaborating in teams, and am comfortable working independently while always looking to grow and contribute more in a dynamic work environment.

EXPERIENCE

Software Engineer (Fulltime) | *Nov 2024 – Present*

PT Transformasi Data Digital

- Maintained and developed client websites focused on top-up systems and digital transaction flows using PHP (Native OOP and Laravel) and MySQL.
- Monitored and troubleshot production transaction systems, investigating API and database issues affecting transaction processing.
- Participated in API development for new client feature integrations and maintained API documentation as requirements and implementations evolved.
- Integrated third-party APIs (Coda, domain provisioning) into client top-up websites as new feature requirements arose.
- Implemented atomic database transactions and decimal-based monetary calculations to maintain financial consistency.
- Collaborated with cross-functional teams to keep communication clear and aligned, establishing structured discussion habits that reduced miscommunication across the team.

Software Engineer (Co-Founder) | *August 2025 – Present*

TokTik Indonesia

- Co-founded TokTik and built the backend using Laravel, Golang, MySQL, and REST APIs for an electricity token and digital top-up platform, serving 2,729+ active users as of November 2026.
- Driven by infrastructure cost efficiency, migrated high-traffic transaction and product API endpoints to Go after exhausting Laravel optimizations (query tuning, caching, and OPcache); load-tested via k6 and implemented request concurrency limits to prevent server overload under peak load.
- Implemented clean architecture with repository pattern, service layer,and raw SQL for the Go service.
- Connected the Laravel and Go services via gRPC for callbacks, implemented token-based authentication middleware, and documented API interactions for system maintenance.
- Integrated payment gateways and digital product aggregators, implementing idempotent callback processing using transaction order IDs and status validation to safely handle asynchronous payment events and prevent duplicate transactions.
- Cached product data and authenticated user profiles in Redis as serialized JSON strings with TTL-based expiry to reduce MySQL reads.
- Implemented automated unit testing in the CI/CD pipeline, running tests before Docker image builds and automated deployment (Alibaba Cloud).
- Built the Agent portal of TokTik with Next.js and integrated it with the Go backend APIs, covering withdrawal and redemption flows.

Software Engineer (Freelance / Project-based) | *Mar 2025 – Aug 2025*

PT UMBI Teknologi Indonesia

- Collaborated on the maintenance, optimization, and development of the Solo Raya Great Sale event website using Laravel and MySQL.
- Responsible for data management, database optimization, and infrastructure stability monitoring.
- Optimized database performance across 23,000+ merchant records, 5 million+ transaction records, and 212,041 users through indexing and query optimization.
- Collaborated within the development team to ensure timely delivery of features, optimizations, and bug fixes.
- Coordinated with admin team to handle user and tenant issue reports rapidly.

Software Engineer (Freelance / Project-based) | *Apr 2024 – Jul 2024*

PT UMBI Teknologi Indonesia

- Maintained and developed the Solo Raya Great Sale event website.
- Updated UI, fixed bugs and errors, and ensured consistent website performance.
- Added new features and adjusted existing ones based on event requirements.

SKILLS

Programming Languages: PHP, Golang, JavaScript, TypeScript, Node.js, HTML5, SQL

Frameworks & Libraries: Laravel, Gin, React.js, Next.js, Vue.js

Databases: MySQL, PostgreSQL

Caching & Event Driven: Redis

DevOps & Infrastructure: Linux, Git, Github, GitHub Actions (CI/CD), Docker, Server Deployment, gRPC, Apache, Nginx, PHP-FPM

Other: Microservices Architecture, REST API, Payment Gateways, Unit Testing, Alibaba Cloud, Agile/Scrum, SDLC

EDUCATION

Bachelor of Information Technology GPA: 3.74

University Duta Bangsa Surakarta | Sep 2020 – Sep 2024

First-authored academic publication in LOFIAN Vol. 4 No. 1 (August 2024): IoT-based household energy monitoring system using ESP32 and PZEM-004T sensors.

LANGUAGES

- Indonesia (Native)
- English (Working Proficiency)

PROJECTS

Portal Nevalis — ERP System

Laravel (Eloquent ORM, Jobs & Queues, Events/Listeners, Middleware, Service Container), ReactJS, MySQL, Ajax, jQuery, REST API, GitHub Actions

- Designed and implemented the end-to-end architecture of an ERP system under PT Transformasi Data Digital.
- Collaborated on technical decisions through team discussions and participated in peer code reviews to maintain code quality and consistency using the SDLC method.
- Implemented SSE for asynchronous background import processing, preventing controller timeout on large data imports.
- Leveraged IndexedDB for client-side draft auto-saving on complex multi-field forms, syncing to server on submission.
- Applied ReactJS selectively on one high-complexity page to manage JavaScript state without overengineering.
- Set up GitHub Actions CI/CD pipeline on shared hosting via SSH for automated deployment.

Protowa — WhatsApp API Platform

React.js, Laravel, Node.js, PostgreSQL, Microservice

- Built a microservice-based application consisting of a React frontend, Laravel backend, and an independent Node.js service for WhatsApp API operations.
- Migrated from whatsapp-web.js (Puppeteer, ~40MB/session) to Baileys (WebSocket-based) for significantly lower resource footprint.
- Solved Baileys credential persistence through iterative approach: moved from impractical file storage to in-memory caching, then to a final solution persisting both ephemeral keys and session data to PostgreSQL, ensuring full session survival across restarts.
- Implemented SSE combined with Node.js Event Emitter for real-time QR scan notifications outside the HTTP request cycle.

Incubator System

Golang, PostgreSQL, Linux, ESP32 (Arduino/C++), Vue.js, TypeScript, Github Actions

- Architected a multi-device IoT backend using Golang to handle high-frequency concurrent requests from multiple sensor devices sending data every second.
- Built Golang backend with PostgreSQL and PgBouncer connection pooling, load tested with k6 achieving 200 req/s on a 4-core 4GB RAM server at ~50% CPU usage.
- Implemented real-time monitoring using Server-Sent Events (SSE) with Vue.js and TypeScript dashboard, enabling live sensor data display.

- Integrated ESP32 microcontrollers communicating directly with the Golang REST API via HTTPS, sending sensor readings every second.

atsin.id — ATS-Friendly CV Generator

Vue.js, TypeScript, Node.js, Playwright, IndexedDB

- Built a privacy-first CV generator using IndexedDB for client-side storage (no login required), personal data never leaves the browser.
- Implemented PDF generation via Node.js and Playwright for single-click download experience.
- Built with Vue.js and TypeScript as a deliberate production learning ground for both technologies.